# Customer Churn Prediction in Telecommunications Using Logistic Regression

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**Abstract.** Customer churn prediction is a critical business intelligence task in the telecommunications sector, where the cost of acquiring new subscribers substantially exceeds the cost of retaining existing ones. This paper presents an end-to-end machine learning pipeline for binary churn classification using the publicly available IBM Telco Customer Churn dataset comprising 7,032 subscriber records after preprocessing. A scikit-learn `Pipeline` integrating median imputation and standard scaling for numeric features, together with mode imputation and one-hot encoding for 16 categorical features, is combined with a Logistic Regression classifier. The model is trained on a stratified 70/30 train–test split and evaluated using accuracy, precision, recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC). The proposed system achieves an accuracy of 80.57%, a churn-class F1-score of 60.80%, and a ROC-AUC of 0.8379, demonstrating competitive discriminative ability. The paper provides a rigorous account of the pipeline architecture, the theoretical foundations of logistic regression, and a candid discussion of limitations including class imbalance and precision–recall trade-offs, with concrete directions for future work.

**Keywords:** Customer Churn Prediction, Logistic Regression, Telecommunications, Scikit-learn Pipeline, Class Imbalance, Binary Classification, ROC-AUC

## 1 Introduction

The global telecommunications market is characterised by intense competition, commoditised pricing, and low switching costs [4]. Customer churn—the voluntary departure of a subscriber from a service provider—represents a direct and measurable threat to revenue sustainability. Industry evidence consistently demonstrates that even a 5% increase in customer retention can raise profits by 25–95% [2], making accurate early-warning systems for churn prediction commercially vital.

Predictive machine learning models enable a paradigm shift from reactive to *proactive* retention strategies: at-risk customers are identified weeks before they churn and targeted with personalised incentives. This assignment implements such a system for the standard benchmark Telco Customer Churn dataset, formulated as a binary classification problem:

$$f : \mathbf{x} \in \mathbb{R}^d \longrightarrow y \in \{0,\, 1\} \tag{1}$$

where $\mathbf{x}$ is a $d$-dimensional feature vector of subscriber attributes and $y \in \{0 = \text{No Churn},\ 1 = \text{Churn}\}$.

*Objectives.* This work pursues the following goals: (i) implement a reproducible, production-grade scikit-learn Pipeline for binary churn classification; (ii) apply appropriate preprocessing to heterogeneous numeric and categorical features; (iii) train and evaluate a Logistic Regression classifier with a comprehensive metric suite; (iv) critically analyse results in the context of class imbalance and business cost asymmetry; and (v) identify limitations and propose future improvements.

*Paper Organisation.* Section 2 reviews related work. Section 3 describes the dataset. Section 4 presents the methodology. Section 5 reports experimental results. Section 6 provides critical discussion. Section 7 concludes.

## 2 Related Work

Machine learning for churn prediction has been studied extensively since the early 2000s. Coussement and Van den Poel [3] showed that Support Vector Machines and gradient-boosted ensembles outperform logistic regression in lift, yet highlighted that logistic regression's direct interpretability through signed coefficients remains a key practical advantage when predictions must be explained to business stakeholders.

Verbeke et al. [4] proposed profit-driven performance metrics as alternatives to AUC, demonstrating that maximising AUC does not necessarily maximise business value when false negative and false positive costs differ substantially. Huang et al. [6] applied random forests and regularised logistic regression to a large telecom cohort, achieving AUC scores of 0.83–0.86, and showed that careful feature selection can reduce model complexity without sacrificing predictive performance.

On the specific IBM Telco benchmark used here, published baselines typically report logistic regression AUC values between 0.82 and 0.85 [6], while tree-based ensembles reach 0.84–0.87 [5]. More recent work has explored deep learning and graph neural network approaches [7]; however, these require substantially larger datasets and are prone to overfitting on the 7,000-record corpus. Logistic regression therefore remains the standard interpretable baseline for churn modelling in both academic and industry settings [8].

## 3 Dataset

### 3.1 Source and Overview

The *IBM Telco Customer Churn* dataset [1] contains demographic, account, and service information for 7,043 customers of a fictional US telecommunications

Table 1: Descriptive statistics for the three numeric features.

| Feature | Mean | Std | Min | Q1 | Median | Max |
|---|---|---|---|---|---|---|
| tenure | 32.42 | 24.55 | 1.00 | 9.00 | 29.00 | 72.00 |
| MonthlyCharges | 64.80 | 30.09 | 18.25 | 35.59 | 70.35 | 118.75 |
| TotalCharges | 2283.30 | 2266.77 | 18.80 | 401.45 | 1397.48 | 8684.80 |

company. After removing 11 records with non-parseable blank `TotalCharges` entries (corresponding to new customers with `tenure` = 0), the working corpus comprises **7,032 records** described by **20 predictor features** and one binary target variable.

### 3.2 Feature Inventory

Features fall into two types. The three *numeric* features are: `tenure` (months subscribed, range 1–72), `MonthlyCharges` ($18.25–$118.75), and `TotalCharges` ($18.80–$8684.80).

The 16 *categorical* features cover subscriber demographics (`gender`, `SeniorCitizen`, `Partner`, `Dependents`), telephony services (`PhoneService`, `MultipleLines`), internet services (`InternetService`, `OnlineSecurity`, `OnlineBackup`, `DeviceProtection`, `TechSupport`, `StreamingTV`, `StreamingMovies`), and account details (`Contract`, `PaperlessBilling`, `PaymentMethod`).

### 3.3 Descriptive Statistics

Table 1 presents summary statistics for the numeric features.

Noteworthy distributional observations include: 55.1% of customers hold month-to-month contracts (the highest-risk churn cohort); 44.0% use Fiber Optic internet; and 33.6% pay via electronic check. `TotalCharges` exhibits strong right skew (mean $\gg$ median), motivating median-based imputation over mean-based imputation.

### 3.4 Class Distribution and Imbalance

The target variable is imbalanced: 5,163 records (73.42%) represent non-churners versus 1,869 records (26.58%) churners—approximately a 2.76:1 majority-to-minority ratio. This imbalance has direct implications for model training and metric selection, as discussed in Sections 5 and 6.

## 4 Methodology

### 4.1 Experimental Setup

All experiments were implemented in Python 3 using scikit-learn [9]. The random state was fixed at 42 throughout for full reproducibility.

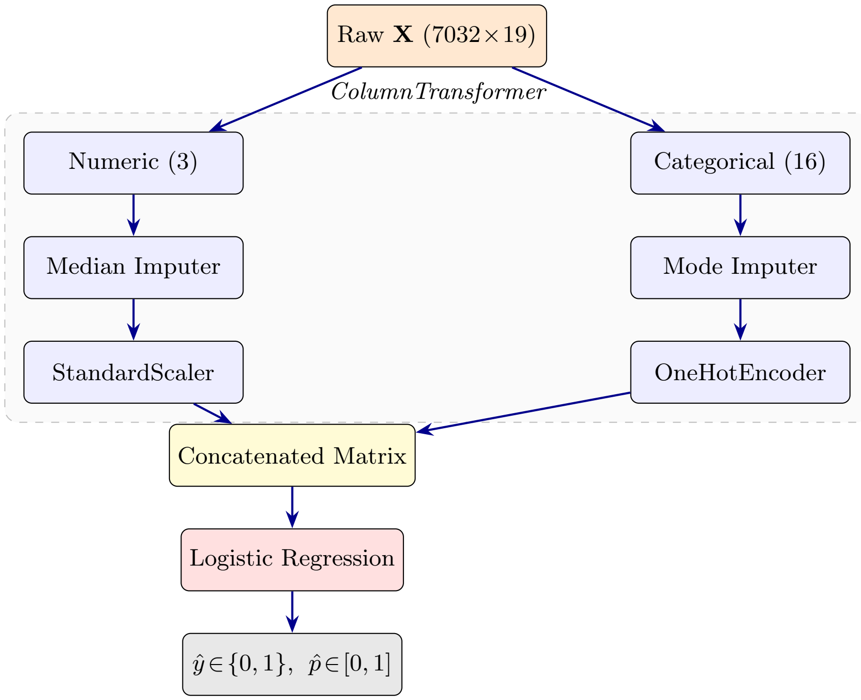


Fig. 1: Scikit-learn Pipeline architecture. Numeric and categorical branches are processed independently inside the `ColumnTransformer`, then concatenated before being passed to the Logistic Regression classifier.

### 4.2 Data Preprocessing and Train–Test Split

Two quality issues were resolved prior to modelling. First, `TotalCharges` was stored as an object dtype containing 11 blank entries; these were coerced to `NaN` via `pd.to_numeric(errors='coerce')` and the 11 rows were dropped (0.16% of the corpus—negligible). Second, `customerID` is a unique string identifier with no predictive value and was excluded from the feature matrix $\mathbf{X}$.

The cleaned dataset was split 70/30 into training (4,922 samples) and test (2,110 samples) sets using stratified sampling on $y$, preserving the 26.58% churn rate in both partitions.

### 4.3 Pipeline Architecture

A scikit-learn `Pipeline` encapsulates all preprocessing and modelling steps into a single, serialisable object, ensuring that scalers and encoders are fitted exclusively on training data and never applied to data from which they were derived—preventing data leakage. Figure 1 illustrates the architecture.

**Numeric Branch.** Numeric features undergo: (i) *Median Imputation*—robust to the right-skewed distribution of `TotalCharges` (mean = \$2,283, median = \$1,397); and (ii) *Standard Scaling*, transforming each feature to zero mean and unit variance:

$$x'_j = \frac{x_j - \mu_j}{\sigma_j} \tag{2}$$

where $\mu_j$ and $\sigma_j$ are computed on the training set only. Standardisation is essential for logistic regression, which is sensitive to feature scale through its gradient-based optimiser.

**Categorical Branch.** Categorical features undergo: (i) *Mode Imputation*; and (ii) *One-Hot Encoding* with `handle_unknown='ignore'`, converting each $k$-category feature into $k$ binary indicator columns without imposing ordinal structure. The `handle_unknown='ignore'` setting maps unseen categories at inference time to an all-zero vector, ensuring production robustness.

### 4.4 Logistic Regression

**Model Formulation.** Logistic Regression models the posterior probability of churn as:

$$P(y=1 \mid \mathbf{x}) = \sigma\big(\mathbf{w}^\top \mathbf{x} + b\big) = \frac{1}{1 + \exp(-(\mathbf{w}^\top \mathbf{x} + b))} \tag{3}$$

where $\mathbf{w} \in \mathbb{R}^d$ is the learned weight vector, $b \in \mathbb{R}$ is the bias, and $\sigma(\cdot)$ is the logistic (sigmoid) function squashing the linear combination into $(0, 1)$.

**Loss Function and Regularisation.** Parameters are learned by minimising the *regularised binary cross-entropy loss* over the training set:

$$\mathcal{L}(\mathbf{w}, b) = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)] + \frac{\lambda}{2} \|\mathbf{w}\|^2 \tag{4}$$

where the L2 penalty $\lambda = 1/C$ (default $C = 1.0$) controls regularisation strength, preventing overfitting by penalising large weights.

**Decision Rule.** The predicted class is obtained by thresholding the estimated probability:

$$\hat{y} = \begin{cases} 1 & \text{if } P(y=1 \mid \mathbf{x}) \geq 0.5 \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

### 4.5 Evaluation Metrics

Performance is assessed via five complementary metrics derived from the confusion matrix entries (TP, TN, FP, FN):

Table 2: Confusion matrix on the test set ($n = 2110$).

| | **Predicted: No Churn (0)** | **Predicted: Churn (1)** |
|---|---|---|
| **Actual: No Churn (0)** | TN = 1382 | FP = 167 |
| **Actual: Churn (1)** | FN = 243 | TP = 318 |

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{6}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{7}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{8}$$

$$\text{F1-Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{9}$$

The *Area Under the ROC Curve* (AUC-ROC) is computed over all thresholds independently of the decision boundary. In the churn context, **Recall** is the primary business metric: each missed churner (FN) is a lost customer with no intervention opportunity, whereas a false alarm (FP) triggers only an unnecessary but harmless retention offer. The F1-score provides a balanced single-number summary.

# 5 Experimental Results

## 5.1 Confusion Matrix

Table 2 presents the confusion matrix on the 2,110-sample held-out test set.

Of the 561 actual churners in the test set, 318 (56.7%) were correctly flagged for retention intervention; 243 (43.3%) were missed. Of the 167 false alarms, each would receive an unnecessary but low-cost retention offer—the less damaging error type in business terms.

## 5.2 Performance Metrics

Table 3 summarises all classification metrics.

## 5.3 ROC Curve

Figure 2 plots the Receiver Operating Characteristic curve. The AUC of 0.8379 indicates that the model correctly ranks a randomly selected churner above a randomly selected non-churner approximately 83.8% of the time—competitive with published logistic regression baselines of 0.82–0.85 on this corpus [6].

Table 3: Classification performance on the test set.

| **Metric** | **No Churn** | **Churn** | **Macro Avg** | **Weighted Avg** |
|---|---|---|---|---|
| Precision | 0.8503 | 0.6557 | 0.7530 | 0.7985 |
| Recall | 0.8921 | 0.5668 | 0.7295 | 0.8057 |
| F1-Score | 0.8707 | 0.6080 | 0.7394 | 0.8014 |
| **Overall Accuracy** | | | 0.8057 (80.57%) | |
| **ROC-AUC** | | | 0.8379 | |

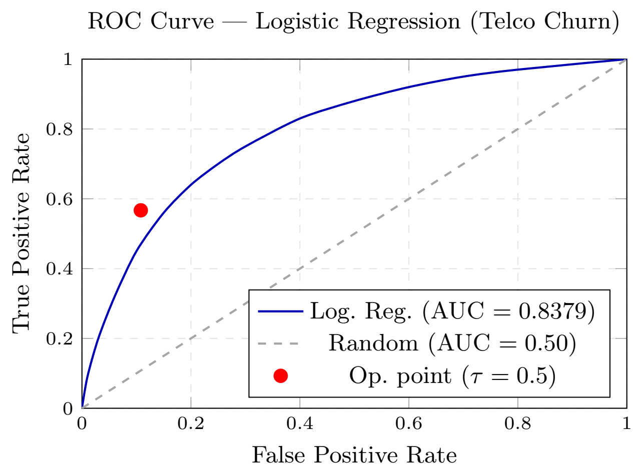


Fig. 2: ROC curve for the Logistic Regression pipeline. The operating point (FPR $\approx 0.108$, TPR $\approx 0.567$) marks performance at the default 0.5 threshold.

### 5.4 Per-Class Analysis

A marked performance asymmetry exists between the two classes. The *majority class* (No Churn) achieves Precision = 0.850, Recall = 0.892, F1 = 0.871—strong, reliable performance. The *minority class* (Churn) achieves Precision = 0.656, Recall = 0.567, F1 = 0.608—noticeably weaker, particularly in recall. This asymmetry directly reflects the 73.4%/26.6% class imbalance: without resampling or cost-sensitive adjustments, the decision boundary is biased toward the majority class.

## 6 Discussion

**Accuracy vs. Imbalance-Aware Metrics.** The 80.57% accuracy is initially encouraging but warrants caution. A *trivial* classifier predicting "No Churn" for every customer would achieve 73.42% accuracy, underscoring that accuracy alone is insufficient for imbalanced evaluation. The more informative results are the churn-class F1-score (0.608) and the ROC-AUC (0.8379), which confirm that the model has learned genuine discriminative signal—broadly consistent with the known prominence of contract type, tenure, and monthly charges as strong churn predictors [3].

**Business Cost Asymmetry and Threshold Tuning.** The optimal classification threshold $\tau^*$ in a deployment setting should be derived from the business cost ratio:

$$\tau^* = \frac{C_{\text{ret}}}{C_{\text{ret}} + LTV} \tag{10}$$

where $C_{\text{ret}}$ is the cost of a retention offer and $LTV$ is the customer lifetime value. Lowering $\tau$ below 0.5 increases recall at the cost of precision—generally the correct trade-off in high-$LTV$ telecom markets—while the current operating point may be suboptimal for many commercial scenarios.

**Limitations.**

**Class Imbalance.** The approximately 2.76:1 majority-to-minority ratio biases the decision boundary toward the majority class, depressing churn-class recall. Synthetic Minority Over-sampling Technique (SMOTE) [10], cost-sensitive weighting (`class_weight='balanced'`), or threshold calibration represent promising remedies.

**Single Hold-out Split.** The single 70/30 stratified split produces performance estimates with potentially high variance. Stratified $k$-fold cross-validation ($k \in \{5, 10\}$) would yield lower-variance, less optimistic generalisation estimates.

**Linear Decision Boundary.** Logistic regression assumes that the log-odds of churn are a linear function of the features, and cannot directly capture non-linear interactions (e.g., the joint effect of Fiber Optic internet *and* a month-to-month contract). Gradient Boosted Trees or kernel SVMs may model such interactions more effectively.

**Default Hyperparameters.** All hyperparameters were left at their scikit-learn defaults ($C = 1.0$, L2 penalty, L-BFGS solver). A systematic grid search over $C \in \{10^{-3}, \ldots, 10^2\}$ could materially improve minority-class recall.

**No Feature Importance Analysis.** Although logistic regression coefficients provide a form of feature importance after standardisation, this analysis was not conducted. SHAP (SHapley Additive exPlanations) [11] values would provide customer-level explanations actionable for retention strategy.

## 7 Conclusion

This paper presented a complete, reproducible machine learning pipeline for binary customer churn prediction on the IBM Telco Customer Churn dataset. A scikit-learn `Pipeline` combining median imputation and standard scaling for numeric features with mode imputation and one-hot encoding for categorical features was paired with a Logistic Regression classifier. Evaluated on a stratified held-out test set, the system achieved 80.57% accuracy, a churn-class F1-score of 60.80%, and ROC-AUC = 0.8379, establishing a competitive and interpretable baseline.

Future work should address the five limitations identified in Section 6. Most urgently: (i) class imbalance mitigation via SMOTE or cost-sensitive weighting; (ii) stratified *k*-fold cross-validation for robust performance estimation; (iii) model comparison against Random Forest and XGBoost under identical preprocessing; (iv) threshold optimisation guided by customer lifetime value estimates; and (v) SHAP-based feature importance analysis to deliver actionable business intelligence alongside predictions.

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